Test Case Register

Final deliverable 7 — SENG 34213 System Development Project

Hotel Website (HW), Forever Hotel. Snapshot: 2026-10-25 (Asia/Colombo).

Assessment status: 525 passed automated cases; 6 HTTP scenarios blocked by DB setup; manual gaps retained

Source revision: `63673ba704b4ae645a8f067e246309b6fb0f03d0`. Branch: `docs/190-release-setup-and-demo-evidence`.

The reports were prepared after this commit; they do not certify a deployed release.

Scope is the HW repository plus identified API Gateway evidence, not the complete multi-subsystem hotel system.

Guideline source: SENG34213_System_Development_Project (1).pdf, section 10.1, printed page 40. The guideline suggests nested testing/security paths; these files are directly under `documents/` as requested by the user.

1. Register conventions

IDs use TC-HW-xx-nn for functional requirements, with SEC/LOAD/DEPLOY supplements. Each case records preconditions, inputs, expected/actual outcome, source, priority, date and status. PASS is limited to the named unit/fixture boundary, never an automatic end-to-end acceptance claim. All rows have high priority for this subsystem handover; critical release gates are discussed in the companion reports.

This PDF contains 30 detailed functional/security/operational cases. The executed commands covered 525 passing automated cases and six setup-blocked HTTP scenarios. Detailed scenario rows and automated assertion totals are related evidence, not additive independent test counts.

2. Run totals

Run	Pass	Failed/blocked	Meaning
Backend unit	431	0	49 isolated unit suites
frontend	50	0	Boundary described in coverage report
gateway	20	0	Boundary described in coverage report
performance	8	0	Boundary described in coverage report
staging	16	0	Boundary described in coverage report
HTTP/security integration	0	6	Hook/database timeout; assertions not reached

3. Detailed functional register

TC-HW-01-01 — Home/public offer metadata

Field	Value
Requirement	HW-01
Type / priority	Frontend / High
Source	frontend/tests/offers.test.mjs
Preconditions	Active/expired synthetic codes
Input / action	Load public offers
Expected	Only eligible public records rendered; no redemption claim
Actual	Passed module assertions; live Manager contract pending
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-02-01 — Hotel navigation/background

Field	Value
Requirement	HW-02
Type / priority	Browser/manual / High
Source	frontend/tests/smoke.test.mjs
Preconditions	Built frontend and approved content
Input / action	Open Home and Contact
Expected	Background on Home; map/text available; no About route
Actual	Not rerun in current report
Status / date	NOT EXECUTED / PENDING / 2026-10-25

TC-HW-03-01 — Catalogue parsing and unsafe images

Field	Value
Requirement	HW-03
Type / priority	Frontend / High
Source	frontend/tests/rooms.test.mjs
Preconditions	Valid and malformed catalogue fixtures
Input / action	Parse prices, images, capacity

Field	Value
Expected	Reject malformed bookable data; reject unsafe image origins
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-04-01 — Compare maximum three room types

Field	Value
Requirement	HW-04
Type / priority	Frontend / High
Source	frontend/tests/comparison.test.mjs
Preconditions	Three selected room IDs
Input / action	Try fourth, remove one, retry
Expected	Fourth blocked until selection removed
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-05-01 — Calendar/date selection rules

Field	Value
Requirement	HW-05
Type / priority	Frontend / High
Source	frontend/tests/availability.test.mjs
Preconditions	Blocked/unavailable/stale fixture dates
Input / action	Select ranges and edit search
Expected	Invalid/stale ranges cannot claim availability
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-06-01 — Password hashing cost and matching

Field	Value
Requirement	HW-06
Type / priority	Backend / High

Field	Value
Source	backend/src/guests/password-hasher.service.spec.ts
Preconditions	Synthetic password and bcrypt hasher
Input / action	Hash and verify correct/incorrect passwords
Expected	bcrypt cost 12, no plaintext hash replacement
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-06-02 — Registration validation

Field	Value
Requirement	HW-06
Type / priority	Backend / High
Source	backend/src/guests/dto/register-guest.dto.spec.ts
Preconditions	Guest DTO fixture
Input / action	Submit invalid names/email/password bounds
Expected	Invalid payload rejected
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-07-01 — JWT signature and expiry

Field	Value
Requirement	HW-07
Type / priority	Backend / High
Source	backend/src/common/utils/jwt-token.service.spec.ts
Preconditions	Synthetic issuer/signing key
Input / action	Verify valid, expired, oversized and altered tokens
Expected	Only signed unexpired guest claims accepted
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-07-02 — Safe return path

Field	Value
Requirement	HW-07
Type / priority	Frontend / High
Source	frontend/tests/auth.test.mjs
Preconditions	Local and external URL samples
Input / action	Validate returnTo
Expected	External/control-character navigation falls back safely
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-08-01 — Booking UI quote/review state

Field	Value
Requirement	HW-08
Type / priority	Frontend / High
Source	frontend/tests/booking.test.mjs
Preconditions	Signed quote fixture, supported one-room selection
Input / action	Review/change/refresh selection
Expected	Stale review invalidated; totals authoritative
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-09-01 — Real two-guest last-room protection

Field	Value
Requirement	HW-09
Type / priority	DB/manual / High
Source	backend/test/integration/booking-race.integration-spec.ts
Preconditions	Disposable localhost CI DB, one physical room, two guests
Input / action	Submit competing reservation transactions
Expected	Exactly one persisted booking and one conflict
Actual	Not rerun; needs disposable CI database

Field	Value
Status / date	NOT EXECUTED / PENDING / 2026-10-25

TC-HW-10-01 — Unsupported checkout promotion

Field	Value
Requirement	HW-10
Type / priority	HTTP integration / High
Source	backend/test/integration/http-consumer-contracts.integration-spec.ts
Preconditions	Real temporary schema and two authenticated guests
Input / action	Submit unsupported promo/price/identity fields
Expected	Reject fields; do not apply invented discounts
Actual	Blocked: database setup timed out
Status / date	BLOCKED / 2026-10-25

TC-HW-11-01 — PayHere signed notification verification

Field	Value
Requirement	HW-11
Type / priority	Backend / High
Source	backend/src/payments/payhere-provider.spec.ts
Preconditions	Synthetic merchant credentials/body
Input / action	Verify valid and tampered provider data
Expected	Bad MAC/amount/environment rejected
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-11-02 — Return URL cannot confirm payment

Field	Value
Requirement	HW-11
Type / priority	Frontend / High
Source	frontend/tests/payment.test.mjs
Preconditions	Unpaid pending status fixture

Field	Value
Input / action	Process redirect/status response
Expected	No confirmed booking from redirect alone
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-11-03 — Provider failure isolation

Field	Value
Requirement	HW-11
Type / priority	Backend / High
Source	backend/src/payments/payhere-payments.service.spec.ts
Preconditions	Mock Retrieval failure
Input / action	Read/reconcile owned payment
Expected	Delayed verification without fabricated payment success
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-12-01 — Email idempotency and sandbox truth

Field	Value
Requirement	HW-12
Type / priority	Backend / High
Source	backend/src/payments/confirmation-email.service.spec.ts
Preconditions	Mock provider and claimed receipt
Input / action	Send/replay/fail/ambiguous outcome
Expected	No repeated accepted receipt; payment remains valid
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-13-01 — Dashboard response contract

Field	Value
Requirement	HW-13

Field	Value
Type / priority	Frontend / High
Source	frontend/tests/dashboard.test.mjs
Preconditions	Owned category/list fixture
Input / action	Parse upcoming/past/cancelled list and totals
Expected	Valid fields only; malformed state rejected
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-13-02 — Cross-guest read/mutation/recovery

Field	Value
Requirement	HW-13
Type / priority	HTTP integration / High
Source	backend/test/integration/http-consumer-contracts.integration-spec.ts
Preconditions	Owner and foreign guest with stored booking
Input / action	Read/modify/cancel/resume/recover other guest booking
Expected	No records or mutation; no foreign provider request
Actual	Blocked: database setup timed out
Status / date	BLOCKED / 2026-10-25

TC-HW-14-01 — Cutoff/revision and policy checks

Field	Value
Requirement	HW-14
Type / priority	Backend / High
Source	backend/src/bookings/booking-management.service.spec.ts
Preconditions	Frozen clock, policy and signed change preview
Input / action	Modify/cancel before/after boundary or with stale revision
Expected	Only eligible owned revision accepted; staff settlement explicit
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-14-02 — Management UI immutable retry

Field	Value
Requirement	HW-14
Type / priority	Frontend / High
Source	frontend/tests/management.test.mjs
Preconditions	Existing attempt key and malformed/failed response
Input / action	Retry uncertain modification/cancellation
Expected	Retain attempt identity; no false verified success
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-15-01 — Facility facts and generated image labels

Field	Value
Requirement	HW-15
Type / priority	Browser/manual / High
Source	frontend/src/features/services/content.ts
Preconditions	Approved content and asset list
Input / action	Inspect facility cards
Expected	Substitute labels visible; no invented operating claims
Actual	Manual owner/visual confirmation pending
Status / date	NOT EXECUTED / PENDING / 2026-10-25

TC-HW-16-01 — Contact delivery feedback

Field	Value
Requirement	HW-16
Type / priority	Frontend / High
Source	frontend/tests/contact.test.mjs
Preconditions	Mock accepted/sandbox/off/error responses
Input / action	Submit enquiry and inspect status
Expected	No inbox delivery claim from sandbox/acceptance
Actual	Passed

Field	Value
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-16-02 — Enquiry throttle and expiry

Field	Value
Requirement	HW-16
Type / priority	Backend / High
Source	backend/src/contact/contact-rate-limit.guard.spec.ts
Preconditions	Synthetic IP and clock
Input / action	Exceed quota and advance window
Expected	429 until expiry; socket fallback supported
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-HW-17-01 — Responsive/axe/keyboard visual check

Field	Value
Requirement	HW-17
Type / priority	Browser/manual / High
Source	frontend/tests/booking.browser.mjs
Preconditions	Production build plus Chrome/Edge fixture runner
Input / action	Run quality at 320, 768, 1440px
Expected	No audit failures/overflow; useful keyboard flow
Actual	Not rerun; historical quality guide retained
Status / date	NOT EXECUTED / PENDING / 2026-10-25

TC-SEC-01 — Secret-free error logging

Field	Value
Requirement	SEC
Type / priority	Backend / High
Source	backend/src/common/filters/all-exceptions.filter.spec.ts
Preconditions	Secret/card/SQL canary in exception

Field	Value
Input / action	Pass unexpected error to filter
Expected	Generic response and event/status-only log
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-SEC-02 — Production SQL logging rejection

Field	Value
Requirement	SEC
Type / priority	Backend / High
Source	backend/src/config/env.validation.spec.ts
Preconditions	Production config with DB_LOGGING=true
Input / action	Validate environment
Expected	Startup refuses sensitive parameter logging
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-LOAD-01 — One winner in load fixture

Field	Value
Requirement	LOAD
Type / priority	Load tooling / High
Source	performance/test/runner.test.mjs
Preconditions	Synthetic HTTP inventory with one slot
Input / action	Run mixed traffic and competing fixture reservations
Expected	One winner/conflict, persisted status and cleanup
Actual	Passed fixture; not real SQL/staging evidence
Status / date	PASS (specified boundary) / 2026-10-25

TC-LOAD-02 — Detect timeout/image/overbooking/cleanup failures

Field	Value
Requirement	LOAD

Field	Value
Type / priority	Load tooling / High
Source	performance/test/runner.test.mjs
Preconditions	Stalled image, failed image, two winners or failed cleanup
Input / action	Run each injected fault mode
Expected	Report fails; timeout or unresolved hold recorded
Actual	Passed
Status / date	PASS (specified boundary) / 2026-10-25

TC-DEPLOY-01 — Failed candidate restores old image

Field	Value
Requirement	DEPLOY
Type / priority	Staging tooling / High
Source	deploy/staging/test/release.test.mjs
Preconditions	Mock current and candidate immutable digests
Input / action	Inject health/smoke failure
Expected	Restore previous pair; release remains failed
Actual	Passed mock; live rollback pending
Status / date	PASS (specified boundary) / 2026-10-25

TC-SEC-03 — Unpaid payment return smoke contract

Field	Value
Requirement	SEC
Type / priority	Staging tooling / High
Source	deploy/staging/test/smoke.test.mjs
Preconditions	Synthetic HTTP login/unpaid booking
Input / action	Submit forged status=success return parameter
Expected	No stay-confirmed output from unpaid state
Actual	Passed synthetic contract
Status / date	PASS (specified boundary) / 2026-10-25

4. Execution limitations and release actions

The integration runner returned six failed tests following a beforeAll database connection timeout; this register preserves the failed runner outcome and labels its cause as environment blocking. No production/shared rows were seeded by these checks. Browser, live PayHere/Brevo, real two-guest SQL contention, 300-user load and live staging rollback must be executed separately. HW-10 online redemption, deposits and deferred controls remain functional gaps; passing validation tests do not implement those features.

5. Reviewer checklist

Confirm case IDs and requirements, rerun blocked cases with a reachable disposable DB, attach exact release CI/artifacts, review manual accessibility/content, and enter reviewer/date through the team process. No supervisor or teammate signature is claimed.

Evidence and sign-off

Current requirement/screen mapping: release traceability.

Prepared from executable results and source inspection; reviewer/supervisor acceptance is not supplied.

Do not attach real environment files, passwords, JWTs, card data or raw provider responses.